

YUCHEN YANG

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PROFESSIONAL EXPERIENCES

Assistant Professor The Pennsylvania State University , PA, United States	2025.07 - Current
Research Assistant for <u>Dr. Yinzhi Cao</u> The Johns Hopkins University , MD, United States	2020.03 - 2025.05
Student Associate for <u>Dr. Shao-yuan Lo</u> , <u>Dr. Kwonjoon Lee</u> , and <u>Dr. Behzad Dariush</u> Honda Research Institute , CA, United States	2023.10 - 2024.02

EDUCATION

Ph.D. The Johns Hopkins University <i>Computer Science</i> Advisor: <u>Dr. Yinzhi Cao</u>	2021 – 2025 MD, United States
M.S. The Johns Hopkins University <i>Security Informatics</i> Advisor: <u>Dr. Yinzhi Cao</u>	2019 – 2021 MD, United States
B.E. Shandong University <i>Software Engineering</i>	2015 – 2019 Shandong, China

AWARDS AND GRANTS

Recipient of Penn State SSRI (Social Science Research Institute) Social AI Award, " <i>Persuasion-Aware Explanations for Safer AI-Mediated Decisions</i> ", PI, \$19,680	2026.03
Recipient of NSF NAIRR Pilot Award, " <i>Certifying Internal Memory in Efficient LLMs: Robustness and Integrity of the KV-Cache</i> ", PI, 15,000 GPU-hours on PSC Bridges-2	2026.01
Recipient of NSF NAIRR Pilot Award, " <i>Detect Scams in the Wild: Towards Realistic Benchmarks and Victim-Centered Evaluation</i> ", PI, \$5,000 OpenAI credits and 6,200 GPU-hours on PSC Bridges-2	2026.01
Recipient of USENIX Security Travel Grant	2025.07
Selected as ICLR Notable Reviewer	2025.05
Academic Scholarship , Shandong University	2018.09

PUBLICATIONS

PEER-REVIEWED CONFERENCE PAPERS

Underlined authors are students under my supervision; * indicates equal contribution.

2026	HijackKV: New Threat in Position-Independent KV Cache Reuse <u>Yichi Zhang</u> , <u>Zhiqi Wang</u> , <u>Huan Zhang</u> , Yuchen Yang <i>To appear in the proceedings of USENIX Security Symposium (Usenix), 2026</i> Acceptance Rate: 12.6%
	Rethinking Forgery Attacks on Semantic Watermarks in Black-Box Settings: A Geometric Distortion Perspective <u>Cheng-Yi Lee</u> , <u>Yichi Zhang</u> , Yuchen Yang , <u>Chun-Shien Lu</u> , <u>Jun-Cheng Chen</u>

To appear in the Proceedings of the International Conference on Machine Learning (ICML), 2026

Jailbreaking Safeguarded Text-to-Image Models via Large Language Models

Zhengyuan Jiang, Yuepeng Hu, **Yuchen Yang**, Yinzhi Cao, Neil Gong

In the Findings of the European Chapter of the Association for Computational Linguistics (EACL), 2026

AdaProb: Efficient Machine Unlearning via Adaptive Probability

Zihao Zhao, **Yuchen Yang**, Anjalie Field, Yinzhi Cao

In the proceedings of International Conference on Learning Representations (ICLR) Workshop DATA-FM, 2026

2025 **CertPHash: Towards Certified Perceptual Hashing via Robust Training**

Yuchen Yang, Qichang Liu, Christopher Brix, Huan Zhang, Yinzhi Cao

In the proceedings of USENIX Security Symposium (Usenix), 2025

2024 **Follow the Rules: Reasoning for Video Anomaly Detection with Large Language Models**

Yuchen Yang, Kwonjoon Lee, Behzad Dariush, Yinzhi Cao, Shao-Yuan Lo

In the Proceedings of the European Conference on Computer Vision (ECCV), 2024

SafeGen: Mitigating Sexually Explicit Content Generation in Text-to-Image Models

Xinfeng Li*, **Yuchen Yang***, Jiangyi Deng*, Chen Yan, Yanjiao Chen, Xiaoyu Ji, Wenyan Xu

In the Proceedings of the ACM Conference on Computer and Communications Security (CCS), 2024

RippleCOT: Amplifying Ripple Effect of Knowledge Editing in Language Models via Chain-of-Thought In-Context Learning

Zihao Zhao, **Yuchen Yang**, Yijian Li, Yinzhi Cao

In the Findings of the Association for Computational Linguistics: Empirical Methods in Natural Language Processing (EMNLP), 2024

SneakyPrompt: Jailbreaking Text-to-image Generative Models

Yuchen Yang, Bo Hui, Haolin Yuan, Neil Gong, Yinzhi Cao

In the Proceedings of the IEEE Symposium on Security and Privacy (S&P), 2024

Media Coverage (Selected): MIT Technology Review [↗](#)

2023 **PrivateFL: Accurate, Differentially Private Federated Learning via Personalized Data Transformation**

Yuchen Yang*, Bo Hui*, Haolin Yuan*, Neil Gong, Yinzhi Cao

In the proceedings of USENIX Security Symposium (Usenix), 2023

Earned all Artifact Badges: Artifacts Available, Artifacts Functional, Results Reproduced

Fortifying Federated Learning against Membership Inference Attacks via Client-level Input Perturbation

Yuchen Yang, Haolin Yuan, Bo Hui, Neil Gong, Neil Fendley, Philippe Burlina, Yinzhi Cao.

In the proceedings of IEEE/IFIP International Conference on Dependable Systems and Networks (DSN), 2023

2022 **Addressing Heterogeneity in Federated Learning via Distributional Transformation**

Haolin Yuan*, Bo Hui*, **Yuchen Yang***, Philippe Burlina, Neil Gong, Yinzhi Cao

In the proceedings of the European Conference on Computer Vision (ECCV), 2022

2021 **Practical Blind Membership Inference Attack via Differential Comparisons**

Bo Hui*, **Yuchen Yang***, Haolin Yuan*, Philippe Burlina, Neil Gong, Yinzhi Cao

In the proceedings of Network & Distributed System Security Symposium (NDSS), 2021

PROFESSIONAL SERVICES

FUNDING PANELIST

Panelist, NSF Future CoRe III

2026

CONFERENCE/JOURNAL REVIEWING

PC, USENIX Security Symposium

2026

European Conference on Computer Systems (EuroSys)	2026
IEEE Symposium on Security and Privacy (S&P)	2026, 2027
ACM Conference on Computer and Communications Security (CCS)	2025, 2026
ACM Workshop on Adaptive and Autonomous Cyber Defense (AACD)	2024
AEC , IEEE/IFIP International Conference on Dependable Systems and Networks (DSN)	2024
Reviewer , International Conference on Learning Representations (ICLR)	2025
International Conference on Computer Vision (ICCV)	2025
IEEE Transactions on Dependable and Secure Computing (TDSC)	2023, 2024
IEEE Transactions on Information Forensics & Security (T-IFS)	2024
External Reviewer , IEEE Symposium on Security and Privacy (S&P)	2025
USENIX Security Symposium	2023, 2024
ACM Conference on Computer and Communications Security (CCS)	2022
ACM ASIA Conference on Computer and Communications Security (ASIACCS)	2024
IEEE Computer Security Foundations Symposium (CSF)	2022, 2024
IEEE International Conference on Distributed Computing Systems (ICDCS)	2022
ORGANIZING AND CHAIRING	
Session Chair , IEEE Workshop on Deep Learning Security and Privacy (DLSP)	2024

ADVISING EXPERIENCE

PH.D. STUDENTS

Yuyang Zhang, the Pennsylvania State University, advisor	2026.08 (incoming)
Yichi Zhang, the Pennsylvania State University, advisor	2025.08 - current
Zhiqi Wang, the Pennsylvania State University, advisor	2025.08 - current

MASTER STUDENTS

Anglin Cai, master student at NWU	2026.03 - current
Zhenxiang Huang, master student at HKU	2026.01 - current
Adila Abudurehman, master student at JHU. Now: Software engineer at Microsoft	2023.10 - 2024.02
Manoj Valeti, master student at JHU. Now: Software engineer at AWS Amazon	2023.10 - 2024.02
Amodini Vardhan, master student at JHU. Now: Security engineer at Twilio	2023.10 - 2024.02
Ashi Garg, master student at JHU. Now: Research associate at HLTCOE	2022.09 - 2023.03

UNDERGRADUATE STUDENTS

Scott Garcia, undergraduate student at Penn State, advisor for Honors Program Thesis	2026.03 - current
Nizhang Li, undergraduate student at MUST	2026.02 - current
Zihao Zhao, undergraduate student at JHU. Now Ph.D. student at JHU	2023.12 - 2024.09
Qichang Liu, undergraduate student at Tsinghua University. Now Ph.D. student at UIUC	2024.06 - 2024.09
Yichen Li, undergraduate student at Beijing University	2024.06 - 2024.09
Anning Li, undergraduate student at UESTC. Now: Master student at CMU	2023.06 - 2023.09

TEACHING EXPERIENCE

Instructor , IST362: Cybersecurity Analytics Studio, the Pennsylvania State University	Spring 2026
Instructor , IST866: Threats and Cybersecurity, the Pennsylvania State University	Fall 2025
Teaching Assistant , EN 601.340/440/640: Web Security, the Johns Hopkins University	Fall 2020, 2022

MEDIA COVERAGE

Text-to-image AI models can be tricked into generating disturbing images, MIT Technology Review 	2023.11
AI art generators can be fooled into making NSFW images, IEEE Spectrum 	2023.11
Nonsense prompts trick AIs into producing NSFW images, Technology Networks 	2023.11
Researchers reveal vulnerabilities in AI models, prompting concerns, Cryptopolitan 	2023.11
AI generators can be tricked into making NSFW content, JHU Engineering School News 	2023.11
SneakyPrompt: Revealing the vulnerabilities of text-to-image AI, The Johns Hopkins News-Letter 	2023.12

PATENT

2025	System and Method Using Reasoning for Video Anomaly Detection with Large Language Models Yuchen Yang , Kwonjoon Lee, Shao-Yuan Lo, Behzad Dariush <i>Under US Patent Application (No.20250259448), 2025</i>
2018	A New System for Stock Volatility Prediction by Using Privileged Support Vector Machines Yuchen Yang , Zheng Yan, Haoyang Li, Zhixuan Lv, Weiwei Zhao, Simiao Zhao <i>Under IP Australia Application (No.2018101304), 2018</i>

INVITED TALKS

"Diagnose, Correct, Steer: Towards Functional and Trustworthy AI Systems", The Pennsylvania State University, Invited talk	2025.3
"Diagnose, Correct, Steer: Towards Functional and Trustworthy AI Systems", Case Western Reserve University, Invited talk	2025.2
"Diagnose, Correct, Steer: Towards Functional and Trustworthy AI Systems", Rice University, Invited talk	2025.2
"Diagnose, Correct, Steer: Towards Functional, Trustworthy AI", Virginia Tech, Invited talk	2025.2
"Diagnose, Correct, Steer: Towards Functional, Trustworthy AI", Rochester Institute of Technology, Invited talk	2025.1
"Diagnose, Correct, Steer: Towards Functional, Trustworthy AI", Monash University, Invited talk	2024.10
"Zero-shot Video Anomaly Detection: Steering LLMs for Rules-based Reasoning", Voxel51, Invited talk	2024.11